MIS171 – Assignment 3 – T2, 2024

Report to the Board of ePowerDynamics

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Introduction

This report presents the findings of a data analysis performed on bike hire demand for ePowerDynamics. The analysis primarily focuses on correlation and regression to determine factors affecting demand. Based on these insights, we provide data-driven recommendations to maximize bike hire demand, with explanations grounded in the results.

SECTION 1 – ANSWER TO QUESTIONS

Q1: EXPLANATION OF THE CORRELATION ANALYSIS

The correlation analysis reveals key relationships between independent variables i.e. Temperature, seasons, weather, etc and bike hire demand. The strongest positive correlation is with temperature (Temp), showing a value of 0.771, indicating that higher temperatures are associated with increased bike hires. The variable with the most negative correlation is quiet periods in rental activities, at -0.776, meaning fewer bikes are hired during quieter periods. Additionally, WindSpeed has a weaker positive correlation (0.137) with bike hire demand, suggesting a minimal relationship. Lastly, variables such as workingday, cloudy, and holiday show relatively low or negligible correlations, indicating they may not have a significant impact on bike hire demand. This analysis helps identify the most and least influential factors in the model.

Q2: EXPLANATION OF THE REGRESSION ANALYSIS

the analysis shows a strong connection between factors like season, weather, and activity levels, and the number of bikes hired. Over 91% of the changes in bike hire demand can be explained by these factors combined, which means they have a significant influence on how many bikes are rented.

Key significant factors affecting bike hire demand include:

1. **Quiet periods**: This has the largest negative impact on bike hires, with a coefficient of -58.055, meaning bike hires significantly decrease during quieter periods.
2. **Modest periods**: Also negatively correlated with bike hires, showing a coefficient of -31.791, suggesting fewer bikes are hired during these times.
3. **Busy periods**: Positively impacts bike hires, with a coefficient of 19.675, indicating that bike demand increases during busier times.
4. **Holidays**: Positively associated with bike hires, with a coefficient of 5.607, although the effect is smaller compared to other factors, rental activities slightly increase during holidays

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The analysis shows that we can be 95% confident that the factors examined significantly impact bike hire demand. This means that the relationships identified such as those between weather conditions, activity levels, and seasonal influences are reliable.

SECTION 2 – RECOMMENDATIONS TO MAXIMISE BIKE HIRE DEMAND

Recommendation 1: Point System & Virtual Games

Explanation of Recommendation 1

To enhance bike hire demand during off-peak seasons, ePowerDynamics could implement a **point system** that Users would earn points for covering certain distances, riding during off-seasons, completing multiple rides within a month, or renting bikes as part of a large group, etc. During winter, when bike rentals tend to decline due to a strong negative correlation of **-0.621**, a **virtual treasure hunt** game could be introduced. Participants would navigate designated trails to collect points, making riding more engaging and rewarding, especially during off-seasons.

Our analysis indicates that **temperature** has a strong positive correlation of **0.771** with bike hires, suggesting that promoting this point system during warmer periods i.e. summer would also encourage greater participation. By gamifying the experience, users are motivated to ride longer distances to earn rewards, increasing rental duration and frequency with friends and family.

Moreover, during busy periods, which correlate with a moderate positive correlation of **0.470**, the point system can attract more riders to popular areas, aligning with higher activity levels. This combination of incentives and fun activities not only encourages users to ride but also enhances overall engagement with the ePowerDynamics platform, leading to increased bike hires year-round.

Recommendation 2: AI Dashboard

PowerDynamics could enhance bike hire demand by integrating an AI-powered dashboard on both electric as well as pedal bikes, which includes GPS tracking, tabs like temperature display, kilometres driven, which can also be connected to user's cell phone via Bluetooth. The AI would suggest trending locations like cafes and tourist spots, appealing to tourists and younger riders. This feature will create an engaging experience and increase rental duration.

Explanation of Recommendation 2

To maximize bike hire demand, ePowerDynamics should implement an AI-driven system that combines GPS navigation and user engagement. By installing GPS trackers on bikes, tourists and teenagers can explore destinations more efficiently. The bike screen could display GPS routes, temperature, and distance travelled, while AI offers real-time suggestions for trending spots like cafes, stores, and tourist attractions, enhancing the overall rider experience. Additionally, users could connect their cell phones via Bluetooth to sync their ride data, receive notifications, and customize their ride experience, making the service even more interactive and convenient.

To capitalize on the increased bike hire demand during holidays (+5.607 additional bikes hired), ePowerDynamics could introduce special holiday-themed events or promotions, encouraging families and groups to rent bikes for festive outings. Additionally, to address the significant drop in bike hires during quieter periods (correlation of -0.776), AI-powered features could connect users with others nearby interested in social rides, fostering a sense of community and stimulating demand. Personalized AI recommendations would further enhance the experience by encouraging spontaneous, longer, and more frequent rides. This dynamic strategy would effectively boost overall bike hire demand.

Conclusion:

In conclusion, this report highlights the significant factors influencing bike hire demand for ePowerDynamics and offers strategic, data-driven recommendations to maximize this demand. By analysing correlations and regression results, we identified key variables such as temperature, quiet periods, and holidays that impact bike hire trends. To address these insights, we recommend implementing a point system with gamification to engage users during off-peak seasons, and an AI-powered dashboard to personalize and enhance user experience. These solutions, supported by our analysis, will not only boost bike hires during busier periods but also stimulate demand during quieter times, driving sustained growth year-round.

Reflection:

In addition to the analysed factors, ePowerDynamics could explore partnerships with businesses, such as hotels and tourist attractions, to create bundled offers that encourage bike rentals. Collaborating with tourism boards to promote bike-friendly routes can enhance visibility and attract more customers. Additionally, incorporating customer feedback mechanisms could provide insights into rider preferences and areas for improvement, fostering a customer-centric approach. Expanding marketing efforts through social media campaigns targeting eco-conscious consumers can also amplify brand awareness and align with sustainable practices, ultimately driving higher bike hire demand and supporting long-term growth for ePowerDynamics.